



Data Communications and Networking

Fourth Edition

Chapter 19

Network Layer: Logical Addressing



19-1 IPv4 ADDRESSES

An **IPv4 address** is a **32-bit** address that **uniquely and universally** defines the connection of a device (for example, a computer or a router) to the Internet. → Network of many smaller networks

Topics discussed in this section:

Address Space

Notations

Classful Addressing

Classless Addressing

Network Address Translation (NAT)



IPv4 Address

- ❑ An IPv4 address is 32 bits long. 4 bytes / 4 octets
- ❑ The IPv4 addresses are unique and universal.
- ❑ The address space of IPv4 is 2^{32} or 4,294,967,296

IPv4 Address

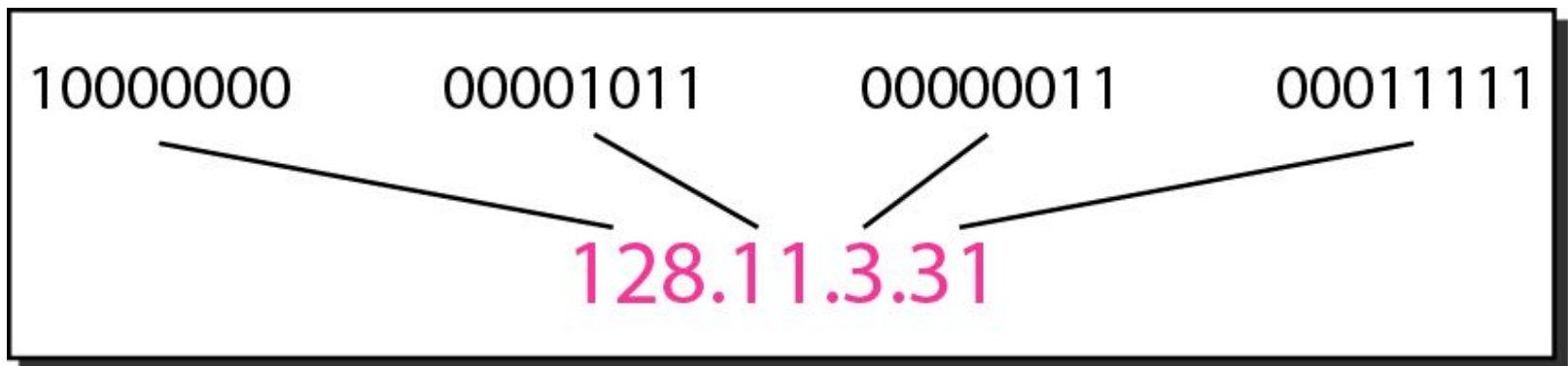
□ Notation

✓ ◆ Binary Notation :

- The IPv4 address is displayed as **32 bits**.
- Each octet is often referred to as a **byte**.

✓ ◆ Dotted-Decimal Notation

- To make the IPv4 address more compact and easier to read, Internet addresses are written in decimal form with a decimal point (dot) separating the byte.



Notation (cont'd)

❑ Hexadecimal Notation

0111 0101 1001 0101 0001 1101 1110 1010

75

95

1D

EA

0x75951DEA

- 8 hexadecimal digits

*- Used in network programming /
socket programming*



Classful Addressing

□ In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

- ◆ If the address is given in binary notation, the first few bits can tell us the class of the address.
- ◆ If the address is given in decimal-dotted notation, the first byte defines the class.

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation



Classful Addressing

Classful Addresses

Unicast Communication - A, B, C Class

(~must be delivered to specific computer)

Multicast Communication - D Class

(~must be delivered to each member of the group)

For reserve - E Class

Classful Addressing

Classes and blocks

- ◆ One problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size.

Table 19.1 *Number of blocks and block size in classful IPv4 addressing*

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved



Classful Addressing

□ Class A addresses were designed for large organizations

◆ The most of the addresses were wasted and were not used.

□ Class B addresses were designed for midsize organizations

◆ Class B is also too large for many organizations.

□ Class C addresses were designed for small organizations

◆ Class C is too small for many organizations.

□ Class D addresses were designed for multicasting

◆ Each addresses in this class is used to define one group of hosts on the Internet.

□ Class E addresses were reserved for future use.

◆ Only a few used, resulting in another waste of addresses.

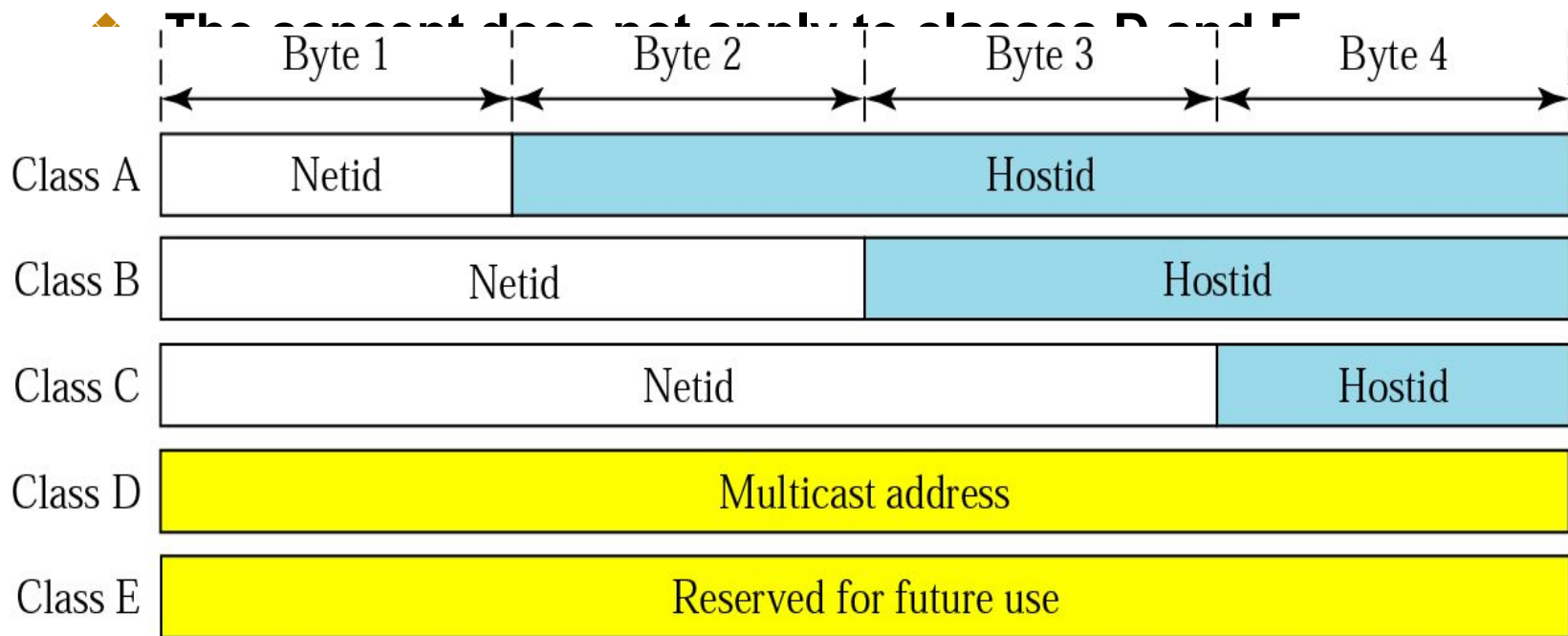
(In classful addressing, a large part of the available addresses were wasted.)



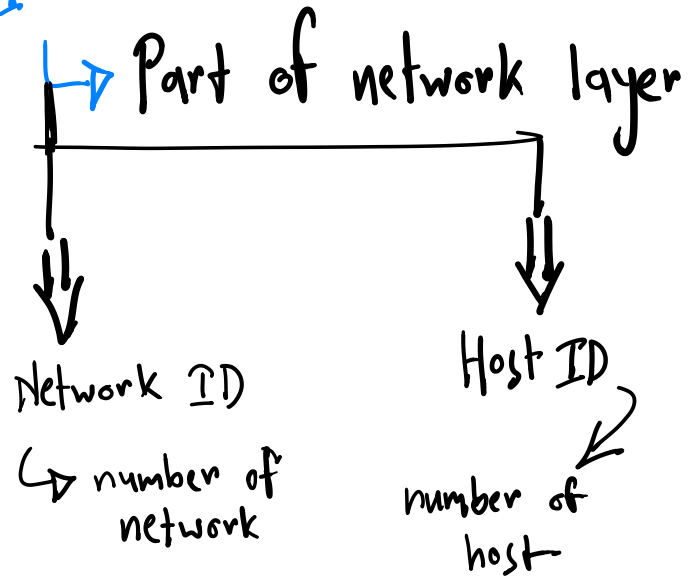
Classful Addressing

□ Netid and Hostid

- ❖ IP address in class A,B, or C is divided into netID and hostID.
 - netID defines a Network, and hostID defines a host in the networks.



IP Address



Class

Octet Value

A

0-127

B

128-191

C

192-223

D

224-239

E

240-255

Subnet Mask:

Class A: 255.0.0.0

Class B: 255.255.0.0

Class C: 255.255.255.0

MASK

separates Network ID & Host ID

□ Mask

- ◆ When a router receives a packet with a destination address, it needs to route the packet.)
- ◆ The routing is based on the network address and subnetwork address.
 - The router outside the organization has a routing table with one column based on the network addresses;
 - The router inside the organization has a routing table based on the subnetwork addresses.
- ◆ The mask is a 32-bit binary number, and the mask can help to find the network and subnetwork address.
 - The routers outside the organization use a Default Mask to find the network address and,
 - The routers inside the organization use a Subnet Mask to find the subnetwork address..



Default Mask

□ Default Mask

- ◆ A default mask is a 32-bit binary number, and the default mask for each class are as follows; 255.0.0.0, 255.255.0.0, 255.255.255.0.
- ◆ Default mask gives the network address when ANDed with an address in the block.
 - If the bit in the mask is 1, the corresponding bit in the address is retained in the output (no change)
 - If the mask is 0, a 0 bit in the output is the result.

Table 19.1 Default masks

Class	Binary	Dotted-Decimal	CIDR
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24



1 Why CIDR Was Introduced (The Problem)

Before CIDR, the Internet used classful addressing:

Class	Range	Default Mask
A	0-127	/8
B	128-191	/16
C	192-223	/24

✗ Problems with Classful Addressing

- Huge IP address wastage
- No flexibility
- Routing tables grew very large

✚ Example waste:

- Need 300 IPs → Class C gives only 254 ✗
- Class B gives 65,534 ✗ (too big!)

➡ CIDR solved this.

2 What is CIDR?

◆ Definition

CIDR is a method of IP addressing that allows **variable-length network prefixes**, independent of IP classes.

◆ Key Idea

- No fixed classes (A, B, C)
- Network size can be any power of 2
- Written as:

CSS

Copy code

IP **address** / prefix length

3 CIDR Notation Explained

Format:

CSS

Copy code

a.b.c.d / n

Where:

- n = number of network bits
- Remaining bits = host bits

✚ Example:

Copy code

192.168.1.10/24

- 24 bits → network
- 8 bits → hosts

4 How CIDR Works (Binary View)

IP address = 32 bits

CIDR splits it into:

CSS

Copy code

[**Network bits**][**Host bits**]

Subnet mask is derived from prefix length.

Subnet Mask

Subnetting

A network is divided into several smaller networks with each subnetwork (or subnet) having its subnetwork address

Subnet Mask :

We change some of the leftmost 0s in the default mask to make a subnet mask.

The number of subnets is determined by the number of extra 1s.

- If the number of extra 1s in n , the number of subnets is 2^n .
- If the number of subnets is N , the number of extra 1s is $\log_2 N$.

	255.255.0.0			
Default Mask	11111111	11111111	00000000	00000000
			16	
	255.255.224.0			
Subnet Mask	11111111	11111111	111	00000000
			3	13

$$2^n = 2^3 = 8 \text{ subnets}$$



VALID SUBNET MASKS

/n	Mask	/n	Mask	/n	Mask	/n	Mask
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

255
254
252
248
240
224
192
128
0

/1: 1000 ... -00
/2: 1100 ... 00
/32: 1111 ... -11

Example 1: 15.200.200.200 $\rightarrow$ Class A

Subnet Mask: 255.0.0.0

Network ID: 15.0.0.0 } zeroed the host id

Broadcast Address: 15.255.255.255 } whenever zero $\rightarrow$ 255

Total number of host bit $\rightarrow$ 3 bytes = 24 bits

Number of hosts = $2^{24} - 2$

$$\hookrightarrow \boxed{2^n - 2} = 16777214$$

Example 2:

192.168.64.1

$\hookrightarrow$ Class C

255.255.255.0

1) Network Address = 192.168.64.0

2) Broadcast Address = 192.168.64.255

3) Total number of host bit = 8 bit

4) Number of hosts = $2^n - 2 = 2^8 - 2 = 254$

Exercise: Given network 172.16.0.0/16, create four equal-sized subnets.

- Determine how many bits to borrow for 4 subnets.
- Write each subnet's network address, subnet mask (in dotted decimal and CIDR), usable host range, and broadcast address.

Solⁿ:

From here, Default mask: /16 $\rightarrow$ 255.255.0.0

Address range: 172.16.0.0 - 172.16.255.255

Now, number of subnets needed = 4

$$2^n \geq 4 \Rightarrow n = 2$$

Borrow 2 bits from the host

$\rightarrow$ /16 + 2 = /18 $\rightarrow$ 255.255.192.0

Now, Block size is determined from the subnet mask

In the 3rd octet: $256 - 192 = 64$

So, each subnet should increase by 64 in the 3rd octet.

4 List all 4 subnets

• Subnet 1

- Network address: 172.16.0.0/18
- Usable host range: 172.16.0.1 - 172.16.63.254
- Broadcast address: 172.16.63.255 ✖

• Subnet 2

- Network address: 172.16.64.0/18
- Usable host range: 172.16.64.1 - 172.16.127.254
- Broadcast address: 172.16.127.255 ✖

• Subnet 3

- Network address: 172.16.128.0/18
- Usable host range: 172.16.128.1 - 172.16.191.254
- Broadcast address: 172.16.191.255 ✖

• Subnet 4

- Network address: 172.16.192.0/18
- Usable host range: 172.16.192.1 - 172.16.255.254
- Broadcast address: 172.16.255.255 ✖

5 Hosts per subnet (extra exam point)

Remaining host bits:

 Copy code

$$32 - 18 = 14 \text{ bits}$$

Hosts per subnet:

$$2^{14} - 2 = 16382 \text{ hosts}$$

✓ Final Summary Table

Subnet	Network Address	Subnet Mask	Usable Host Range	Broadcast 
1	172.16.0.0/18	255.255.192.0	172.16.0.1 - 172.16.63.254	172.16.63.255
2	172.16.64.0/18	255.255.192.0	172.16.64.1 - 172.16.127.254	172.16.127.255
3	172.16.128.0/18	255.255.192.0	172.16.128.1 - 172.16.191.254	172.16.191.255
4	172.16.192.0/18	255.255.192.0	172.16.192.1 - 172.16.255.254	172.16.255.255

Q1 -> How many subnets are created if you borrow 3 bits?

Q2 -> How many usable hosts per subnet in a /27 network?

Q3 -> What is the block size for subnet mask 255.255.255.224?

Q4 -> Which octet does subnetting occur in for /20?

Q5 -> How many host bits are there in a /23 network?

Solⁿ:

Q1 $\rightarrow$ 3 bits $\rightarrow$ subnets that can be created $= 2^3 = 8$

Q2 $\rightarrow$ /27 $\rightarrow$ Host bits, $32 - 27 = 5 \rightarrow$ Usable host $= 2^5 - 2 = 30$

Q3 $\rightarrow$ Block size $= 256 - 224 = 32$

Q4 $\rightarrow$ /20 $\rightarrow$ falls between /16 and /24 $\rightarrow$ 3rd Octet

Q5 $\rightarrow$ /23 $\rightarrow$ Host bits, $32 - 23 = 9$

Q6 -> What is the subnet mask for /26 (dotted decimal)?

Q7 -> How many usable hosts in a /26 subnet?

Q8 -> What is the block size for /22?

Q9 -> Given IP 192.168.10.77/27, what is the network address?

Q10 -> How many subnets are created from /24 -> /27?

Solⁿ:

Q6 -> /26 -> 255.255.255.192

Q7 -> /26 -> Host Bits = $32 - 26 = 6$ -> $2^6 - 2 = 62$

Q8 -> /22 -> Subnetting in 3rd octet -> 255.255.252.0 -> $256 - 252 = 4$

Q9 -> 192.168.10.77/27 -> /27 -> 255.255.255.224

∴ Block Size = $256 - 224 = 32$ -> Subnet ranges ↓

0 - 31

32 - 63

64 - 95 *

∴ Network Address -> 192.168.10.64

Q10 -> Borrowed Bits = $27 - 24 = 3$

Subnets = $2^3 = 8$



Supernetting and Address depletion

Supernetting

- Combining several class C addresses to create a larger range of addresses

Address Depletion

- The fast growth of the Internet led to the near depletion of the available addresses.
- Classful addressing, which is almost obsolete, is replaced with classless addressing.



Network Address

□ Network Addresses

- ◆ The first address in a block is normally not assigned to any device;
- ◆ It is used as the network address that represents the organization to the rest of the world.

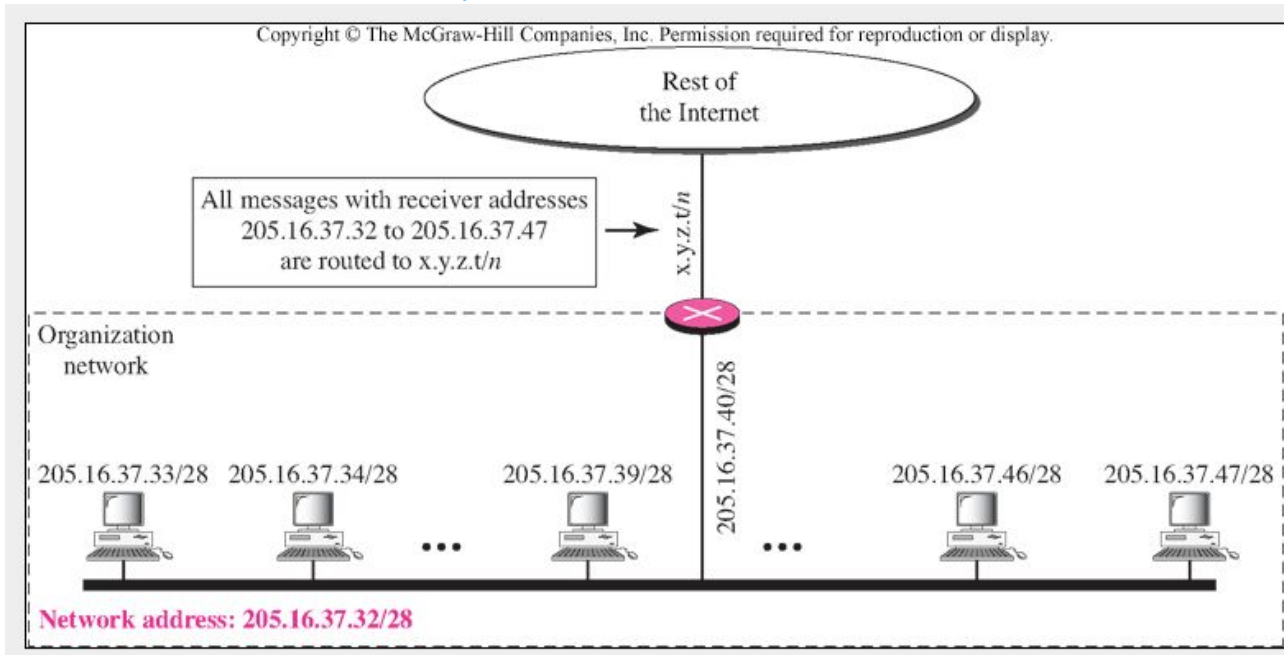


Figure 19.4 *A network configuration for the block 205.16.37.32/28*

Network Address

□ Hierarchy

✓ IP addresses have levels of hierarchy.

For example, a telephone network has three levels of hierarchy.

- The leftmost 3 digits define the area code, the next 3 digits define the exchange, the last 4 digits define the connection of the local loop to the central office.

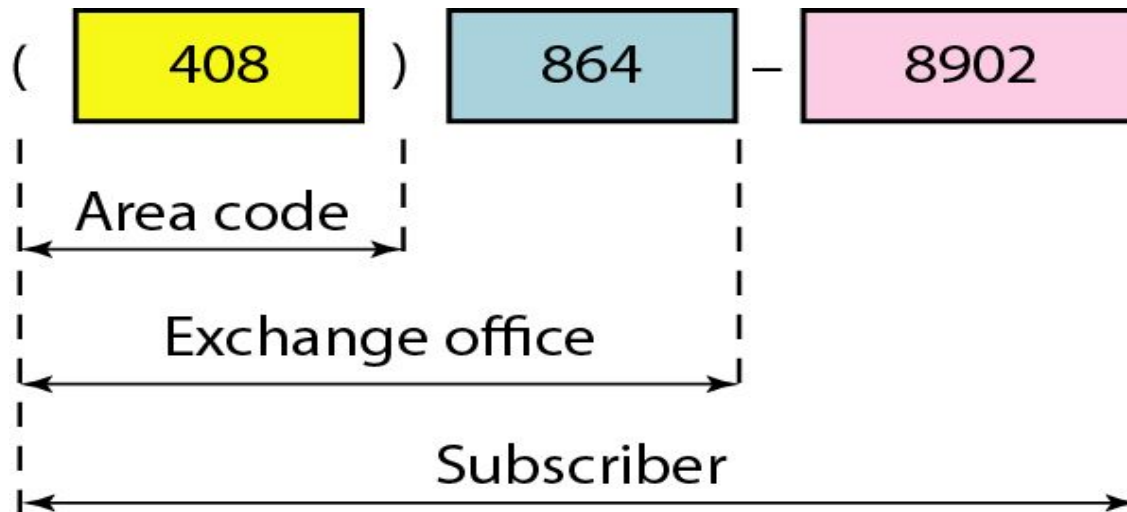


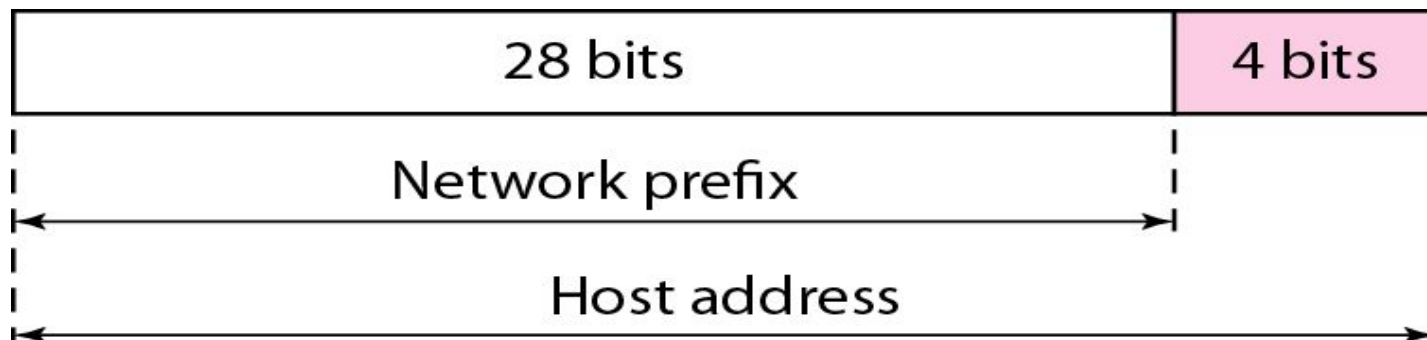
Figure 19.5 *Two levels of hierarchy in an IPv4 address*

Subnetting and Supernetting

❑ Two-level Hierarchy : No Subnetting

- ❖ Each IP address in the block can define only two-level of hierarchy when not subnetted.
 - the leftmost n bits (prefix) define the network;
 - the rightmost $32 - n$ bits define the host.
- ❖ The part of the address that defines the network is called the Prefix;
- ❖ The part that defines the host is called the Suffix.
- ❖ The prefix is common to all addresses in the network; the suffix changes from one device to another.)*

Figure 19.6 *A frame in a character-oriented protocol*



Subnetting and Supernetting

Three-Levels of Hierarchy : Subnetting

- ◆ Creating clusters of networks (called subnets)

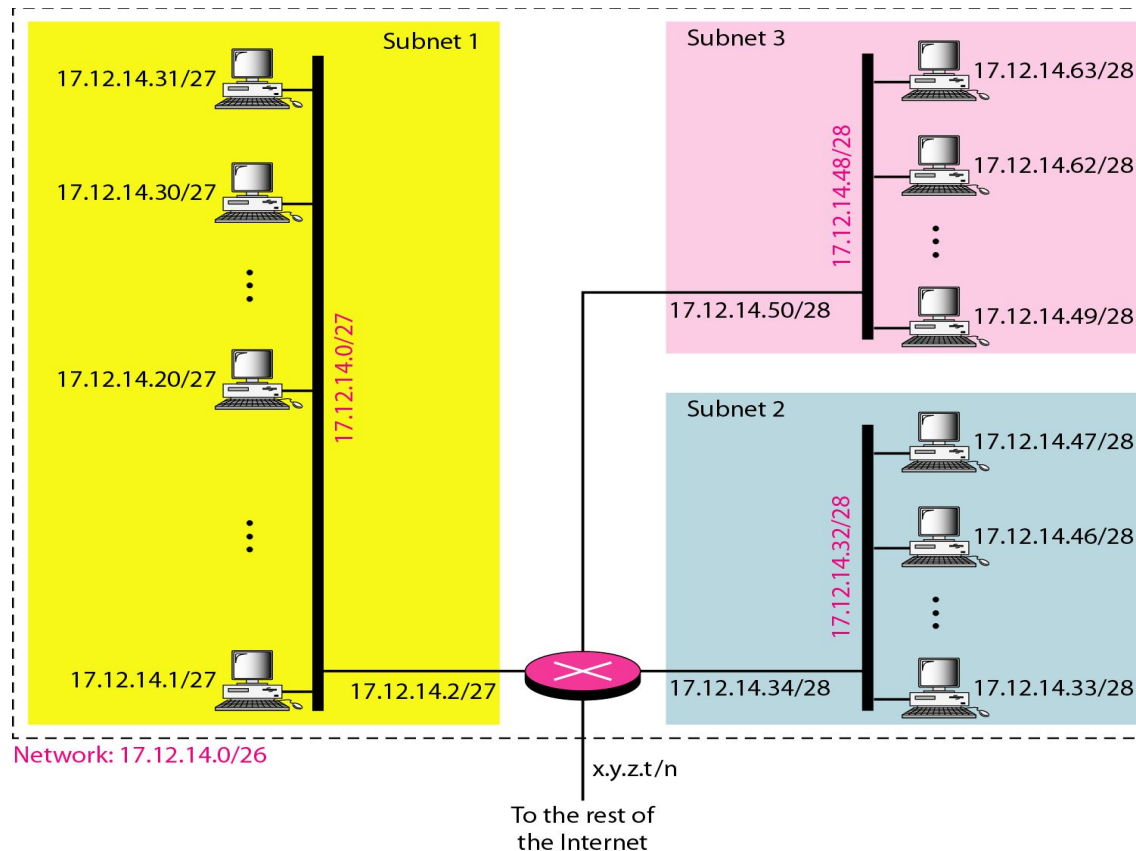


Figure 19.7 Configuration and addresses in a subnetted network

Subnetting and Supernetting

□ We have three levels of hierarchy through subnetting.

◆ The subnet prefix length can differ for the subnets.

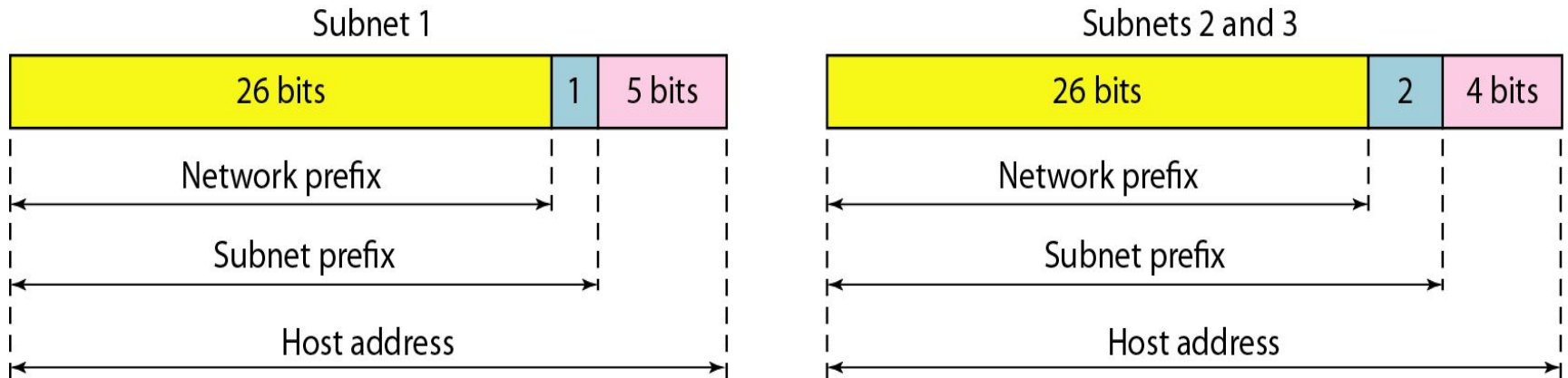
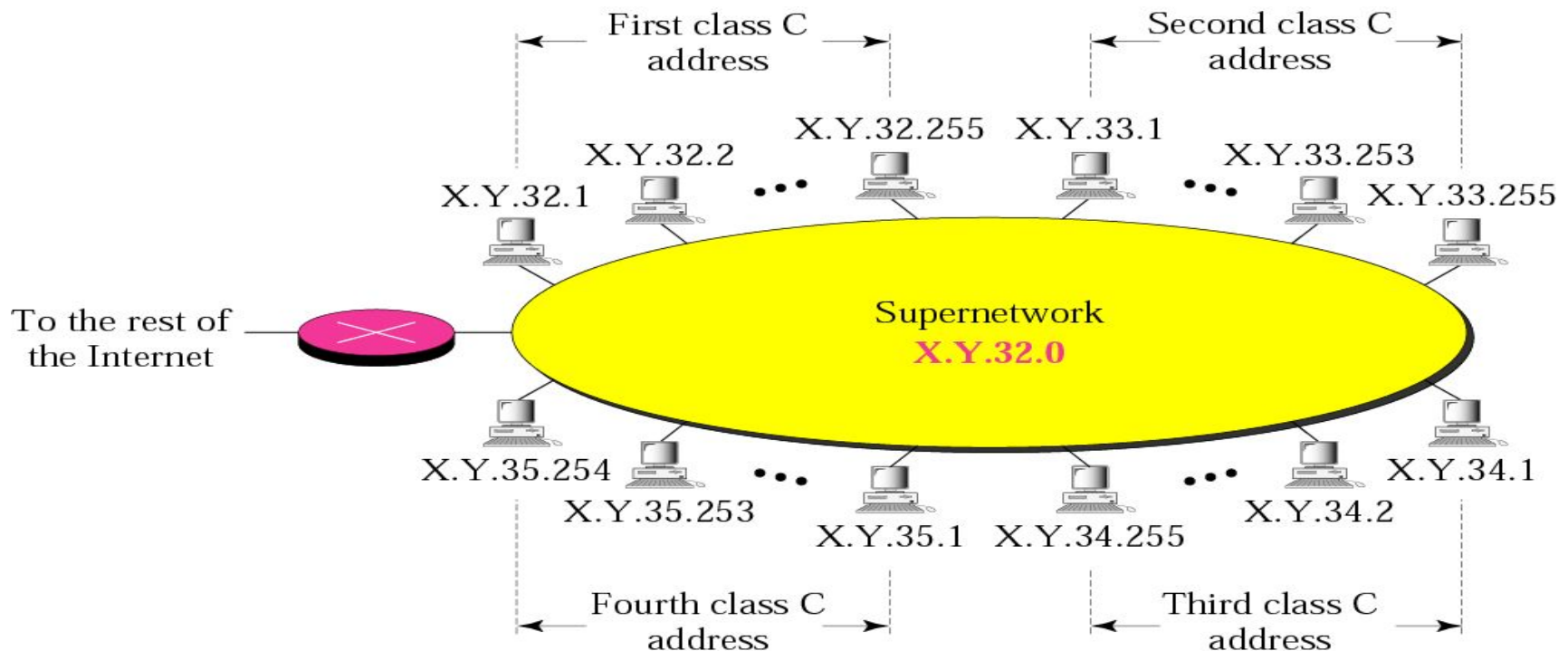


Figure 19.8 *Three-level hierarchy in an IPv4 address*

Supernetting

❏ Supernetting

- ❖ A maximum number of Class C is 256 addresses,
- ❖ If organization needed more addresses, The Supernetting can combine several class C blocks to create a larger range of addresses.
(The mask changes from /24 to /22)



1 Why NAT Is Needed

The core problem:

- IPv4 has limited addresses
- Not enough public IPs for all devices

The solution:

Use private IP addresses internally and translate them to public IPs at the network edge.

This translation is done by NAT.

2 What Is NAT?

Definition:

NAT is a technique where a router modifies the IP address (and sometimes port number) in packet headers as packets pass between private and public networks.

3 Where NAT Happens

• NAT is usually implemented at:

- Border router
- Gateway firewall
- Home router

vbnet

 Copy code

Private Network → [NAT Router] → Public Internet



Network Address Translation (NAT)

❑ Network Address Translation (NAT)

- ◆ NAT enables a user to have a large set of address internally and one address, or a small set of addresses, externally.
- ◆ The Internet authorities have reserved 3 sets of addresses as private addresses.
 - Any organization can use an address out of this set without permission from the Internet authorities.
 - They are unique inside the organization, but they are not unique globally.
 - No router will forward this packet as the destination address.

Table 19.3 *Addresses for private networks*

Range			Total
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}



Network Address Translation (NAT)

□ NAT Implementation

- ◆ The router that connects the network to the global address uses one private address and one global address.
- ◆ The private network is transparent to the rest of the Internet; the rest of the Internet sees only the NAT router with the address 200.24.5.8.

Site using private addresses

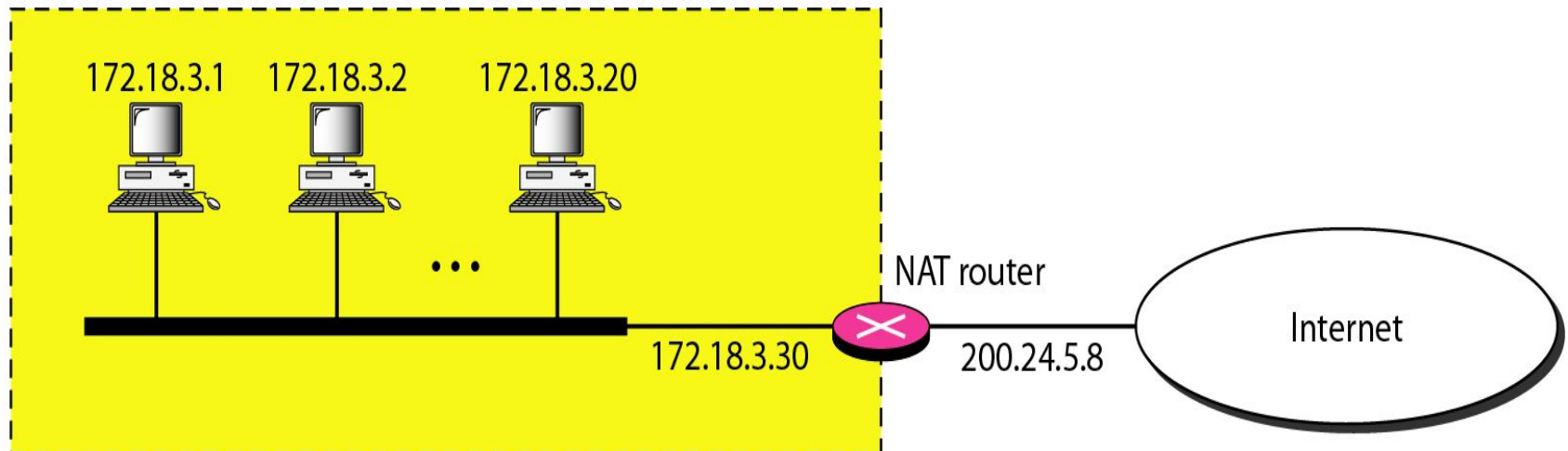


Figure 19.10 *A NAT implementation*

Network Address Translation (NAT)

Address translation

- ◆ All the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address.
- ◆ All incoming packets also pass through the NAT router, which replaces the destination addresses in the packet with the appropriate private address.

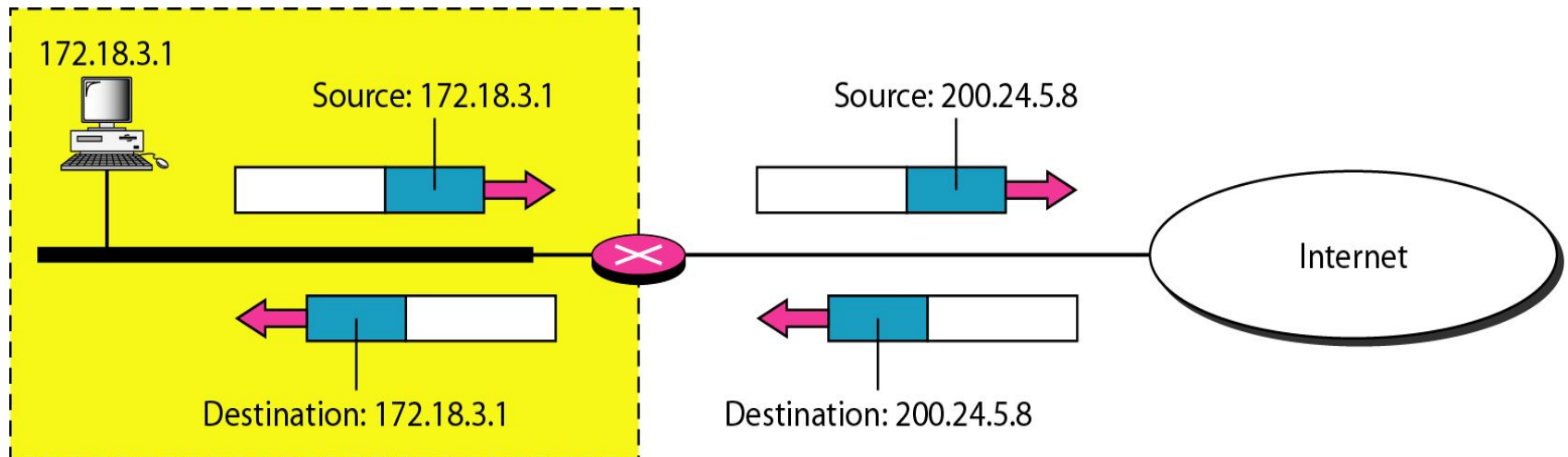


Figure 19.11 *Addresses in a NAT*

Network Address Translation (NAT)

Translation Table

- ◆ When the router translates the source address of the outgoing packet, it also makes note of the destination address – where the packet is going.
- ◆ When the response comes back from the destination, the router uses the source address of the packet to find the private address of the packet.

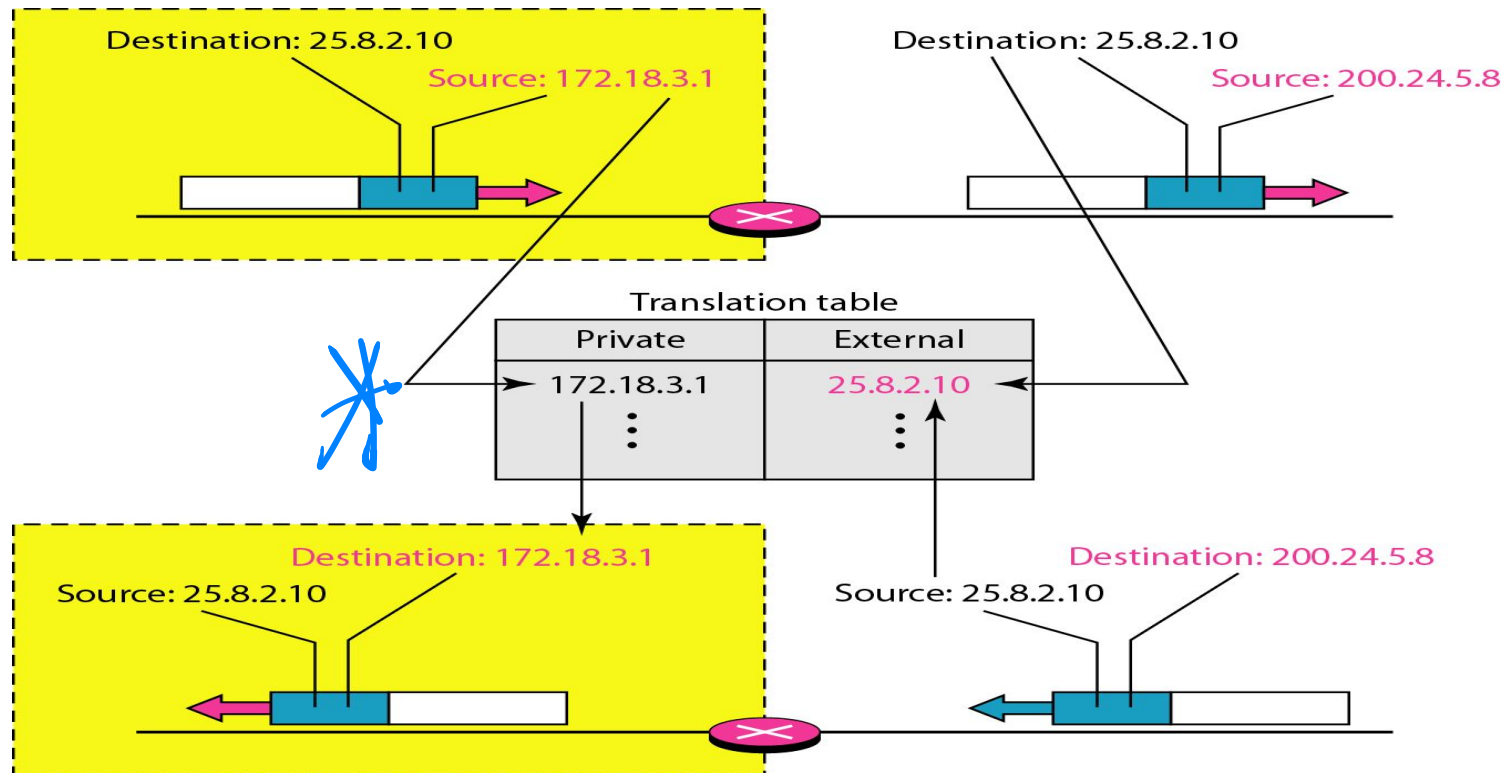


Figure 19.12 NAT address translation

Network Address Translation (NAT)

□ Using both IP addresses and port numbers

Table 19.4 *Five-column translation table*

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...



19-2 IPv6 ADDRESSES

Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.

Topics discussed in this section:

Structure

Address Space



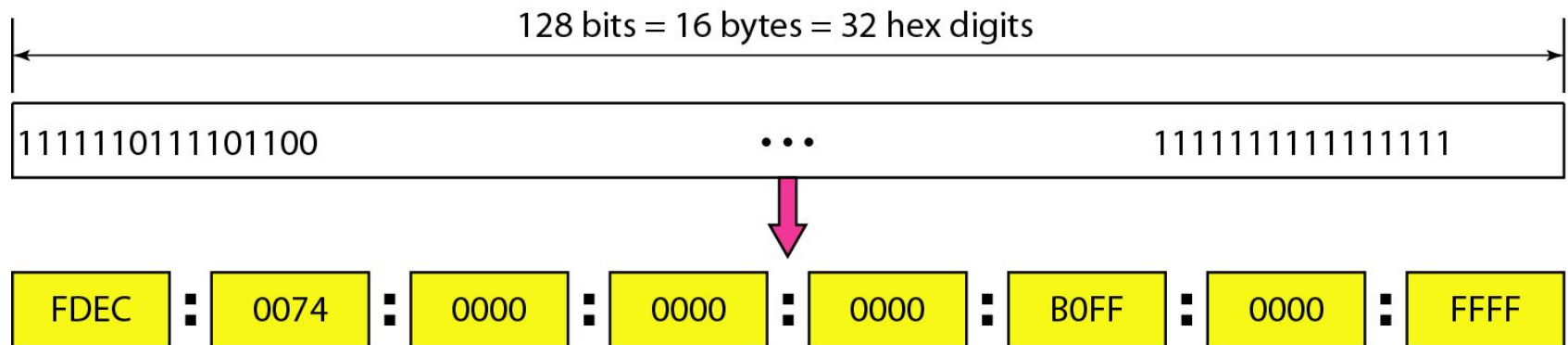
Structure - IPv6

❑ An IPv6 address consists of 16 bytes (Octets); it is 128 bits long.

❑ Hexadeximal Colon Notation

❖ In this notation, 128 bits is divided into eight sections, each 2 bytes in length.

❖ Therefore, the address consists of 32 hexadecimal digits, with every four digits separated by a colon.



Abbreviation

- ❑ Although the IP address, even in hexadecimal format, is very long, many of the digits are zeros.
- ❑ The leading zeros of a section (four digits between two colons) can be omitted.
 - ◆ Only the leading zeros can be dropped, not the trailing zeros.

Original

FDEC : 0074 : 0000 : 0000 : 0000 : B0FF : 0000 : FFF0

Abbreviated

FDEC : 74 : 0 : 0 : 0 : B0FF : 0 : FFF0

More abbreviated

FDEC : 74 : : B0FF : 0 : FFF0

Gap



Address Space

- IPv6 has a much larger address space; 2^{128} addresses are available.



Summary (1)

- ❑ At the Network layer, a global identification system that uniquely identifies every host and router is necessary for delivery of packet from host to host.
- ❑ An IPv4 address is 32 bits long and uniquely and universally defines a host or router on the Internet.
- ❑ In classful addressing, the portion of the IP address that identifies the network is called the netid.
- ❑ In classful addressing, the portion of the IP address that identifies the host or router on the network is called the hosted.
- ❑ An IP address defines a device's connection to a network.
- ❑ There are five classes in IPv4 addresses. Classes A, B, and C differ in the number of hosts allowed per network. Class D is for multicasting and Class E is reserved.

Summary(2)

- ❑ The class of an address is easily determined by examination of the first byte.
- ❑ Addresses in classes A, B, or C are mostly used for unicast communication.
- ❑ Address in class D are used for multicast communication.
- ❑ Subnetting divides a large network into several smaller ones, adding an intermediate level of hierarchy in IP addressing.
- ❑ Supernetting combines several networks into one large one.
- ❑ In classless addressing, we can divide the address space into variable-length blocks.